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Face Video Steganography for Privacy-protection Automatic Depression Assessment

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Abstract

Video-based automatic depression assessment (ADA) system have been widely explored and deployed for real-world usage. While face videos recorded for ADA contains sensitive and private contents, existing standard depression face video delivery and storage strategies lack effective privacy protection mechanism. To address this, this paper proposes the first depression face video steganography strategy which enables the invertible concealment of sensitive depression face videos, while ensuring user privacy. We decompose the facial secret video into individual frames and hide each secret frame into a cover image of the same size, enabling high-capacity steganography for video data. Meanwhile, we leverage Stable Diffusion to allow users to generate their own cover videos, enhancing controllability. During the hiding process, we propose a novel Invertible Network to ensure the precise recovery of facial attributes and behavioral information. Experimental results demonstrate that our video steganography method largely preserves the accuracy of the assessment of depression from facial videos. Our code will be provided upon acceptance.

1. Introduction

Depression is a prevalent mental disorder that affects more than 300 million people worldwide [9]. With the continuous advancement of machine learning, facial video-based automatic depression assessment (ADA) methods [6, 14, 15, 17, 18, 22, 23, 29–31] that infer depression from human facial behaviour dynamics have been widely explored. Despite the promising prospects of face video-based ADA in real-world applications, such face videos typically contain highly sensitive personal information (e.g., identity, age and gender) However, existing mainstream mechanisms for face video transmission and storage offer limited privacy protections, posing a significant risk of sensitive mental health-related behavioral data leakage. Therefore, there is an urgent need for more secure depression face video protection strategy, where face video steganography that allows sensitive facial information to be embedded into visually plausible non-face videos and transmitted covertly [34], offers a novel perspective and a practical solution.

In recent years, benchmark datasets have played a crucial role in advancing Automatic Depression Assessment (ADA) from facial videos. Early datasets such as AVEC 2013 [26], AVEC 2014 [27], VLOG [1], and CMDC [36] provided raw or lightly processed face videos. While human facial behaviours contain rich cues in reflecting human emotional and affect status [3, 11, 12], these videos pose serious privacy risks due to their identity-revealing nature. To mitigate this, later datasets like AVEC 2016 [28], 2017 [19], and 2019 [20] avoid releasing raw videos and instead offer extracted facial features such as AUs, head poses, and expressions. Although this reduces privacy exposure, it comes at the cost of losing contextual and emotional information critical for accurate depression analysis. In industry, raw patient videos are still commonly recorded for telehealth or mental health screening, often without strong privacy-preserving protocols for sharing or storage [7]. These limitations highlight the urgent need for a secure, information-preserving video sharing method. Motivated by this, we propose a novel steganographic framework tailored for ADA, which enables effective analysis while safeguarding sensitive facial data.

Traditional image steganography methods [2, 8] can only

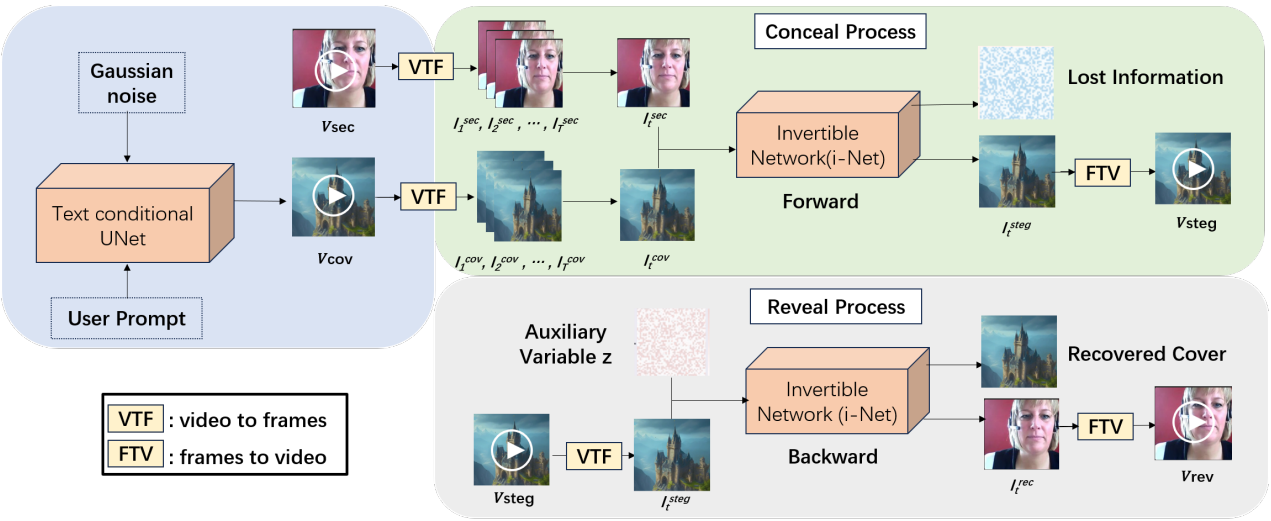

Figure 1. The pipeline of our face video steganography (called DepFaceSteg (DFS)). **(1) Blue part:** the secret face video is first decomposed into a sequence of frames, while a text-guided diffusion model generates an unrelated cover video that matches the resolution, frame rate and length of the secret face video from a user-customized text prompt. **(2) Green part:** Each secret frame is embedded into its corresponding cover frame via an invertible steganography network, resulting in a series of stego-frames. These frames are assembled into the final stego-video for secure transmission and storage; and **(3) Gray part:** During the reveal stage, the stego frames, along with an auxiliary variable z sampled from a Gaussian distribution, are decoded using the same invertible network to accurately reconstruct the original secret video.

embed a small amount of information, which fails to meet the high-capacity requirements of the depression face video hiding task. In recent years, deep learning-based steganography [13, 16, 24, 32] has shown remarkable potential in addressing this limitation by enabling the hiding of an entire image within another image of the same size, using end-to-end trained neural networks, where Generative Adversial Network (GAN) [25] and Diffusion [33, 35] have been widely extended. Despite these advancements, directly applying such methods for depression face video hiding remains challenging as existing face-based ADA approaches typically predict depression status based on subtle facial behavioural cues expressed by the target subjects, i.e. this requires steganography methods to ensure the recovered face videos to have the same subtle facial behaviours as their original face videos (*called secret videos in this paper*). On the other hand, relying on fixed or unrelated cover videos limits practical deployment. In real-world scenarios, users often need control over the visual appearance of the cover video (e.g., contents, background, or style) to ensure consistency with other shared content. To achieve this, we leverage a text-guided diffusion model (e.g., Stable Diffusion) [21] to generate customized cover videos with controllable semantics, enabling flexible and context-aware depression face video steganography.

In this paper, we propose the pioneering, simple but effective depression face video steganography framework (called **DepFaceSteg (DFS)**) encoding all frames in the secret face video of an arbitrary length into another unrelevant non-face video containing the same of more frames, which can be almost fully re-constructed at the reveal stage. In the face video steganography stage, our DFS starts with automatically generating a customized cover video that matches the target resolution, length, and desired appearance using a text-conditioned diffusion model, allowing flexible and user-defined visual control. Then, it decomposes all face frames from the secret face video, and involves an invertible network to encode them into the synthesised cover video, allowing high-capacity hiding while preserving fine-grained facial behaviors crucial for depression analysis. In the reveal stage, the same invertible network is employed to accurately reconstruct the original secret video frames from the stego video without any auxiliary information, preserving not only the pixel-level fidelity but also subtle facial attributes and behavioral cues essential for downstream depression assessment tasks. The entire framework of our approach is illustrated in Fig. 1. The main contributions of this paper can be summarized as follows:

- We propose the first depression face video steganography framework to enhance user privacy protection for face video-based ADA task, which enables the generation of customized cover videos for effective face video hiding.
- Experimental results confirms the effectiveness of our pioneering face video steganography approach in securing user face video data transmission, suggesting that our approach allows not only effective face video hiding but also recovering high-quality face video (evidenced by almost the same ADA results achieved by original and recovered

face videos).

2. Methodology

Overview: This section presents our pioneering face video steganography framework DFS for the task of automatic depression assessment (ADA). As illustrated in Fig. 1, the secret face video to be hidden (denoted as V_{sec}) is first decomposed into a sequence of frames $I_1^{\text{sec}}, I_2^{\text{sec}}, \cdots, I_T^{\text{sec}}$. Meanwhile, a visually irrelevant cover video (denoted as V_{cov}) that have the same resolution and duration as V_{sec} is automatically synthesised via a text-conditioned diffusion model (denoted as Diff), offering controllable semantics for the synthesised cover video, such as background, object motions, and style as:

$$V^{\text{cov}} = \text{Diff}(\text{TC}) \tag{1}$$

where TC denotes the input textual prompt controlling the cover video generation. Each secret frame I_t^{sec} is then paired with its corresponding cover frame $I_t^{\text{cov}} \in V_{\text{cov}}$, and processed by our invertible image steganography network (i-Net), resulting in a stego-image I_t^{steg} that include the secret facial frame I_t^{sec} as:

$$I_t^{\text{steg}} = \text{i-Net}(I_t^{\text{sec}}, I_t^{\text{cov}}) \tag{2}$$

These stego-images are subsequently assembled as the stego-video V_{steg}, ensuring the smoothness and continuity among frames. At the reveal stage, the V_{steg} is split into individual frames, where is processed by the decoder of our to recover the sequence of secret video frames $I_1^{\text{rev}}, \cdots, I_T^{\text{rev}}$, ultimately reconstructing V^{rev} that is expected to be almost the same as the original face video V^{sec}.

2.1. Controllable cover video generation

To enhance the imperceptibility, flexibility, and security of the steganographic process, we extend a diffusion-based video generation model — AnimateDiff [5]. Unlike fixed cover videos, which may introduce semantic or structural correlations that risk exposing hidden content, using generative diffusion models allows us to synthesize cover videos that are both visually realistic and semantically irrelevant to facial data, significantly improving the security of the hidden transmission. Users can specify the desired content of the generated video using natural language prompts, such as "a sunset beach with waves" or "a futuristic city in anime style." AnimateDiff encodes the prompt and generates a video sequence that matches the number of frames and spatial resolution of the target face video to be concealed, serving as V_{cov}. This generation approach offers several advantages:

- **Controllability**: Users have full flexibility to define the video style and content.

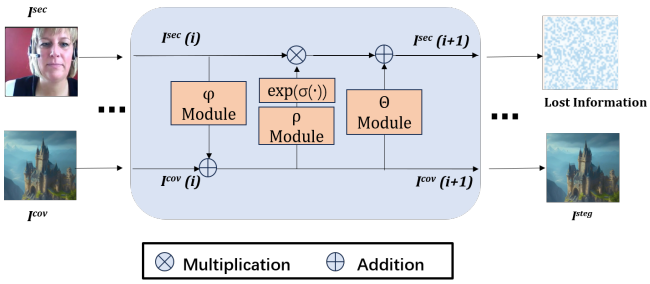

Figure 2. The architecture of our invertible i-Net.

- **High Imperceptibility**: The generated video is semantically unrelated to facial data, making it difficult to infer the hidden content.
- **Structural Alignment**: Techniques such as motion guidance can be employed to ensure that the generated video aligns with the secret video in both dimensions and temporal rhythm.

In practice, we utilize the DDIM sampler to accelerate the generation process, and apply frame alignment to the output video to ensure consistency with V_{sec} in terms of resolution and frame count. which not only preserves the visual realism of the generated cover video but also securely hides the depression-relevant facial signals, thereby enabling privacy-preserving video transmission and storage.

2.2. Invertible network for face video steganography and recover

The key challenge of depression face video steganography lies in embedding and restoring fine-grained facial behavioural information (e.g., micro-expressions, gaze, and posture) for ADA while maintaining high visual quality and imperceptibility. This is particularly critical in applications like ADA, where even subtle distortions may compromise downstream performance. To address this, we design a frame-level steganographic module based on invertible network architectures, enabling lossless and high-capacity encoding of facial frames into unrelated cover videos. Our method is built upon on a Invertible Network, which allows for bijective mappings between the input and output spaces. This means the same model can be used both for hiding and revealing, without requiring any side information.

Network architecture: As illustrated in Fig. 2, our i-Net employs a set of invertible conceal and reveal blocks, which share the same network parameters but operate in opposite information flow directions. The i-th hiding block in the forward process takes $I^{\text{cov}}(i)$ and $I^{\text{sec}}(i)$ as the input, and outputs $I^{\text{cov}}(i+1)$ and $I^{\text{sec}}(i+1)$ as:

$$\begin{aligned} I^{\text{cov}}(i+1) &= I^{\text{cov}}(i) + \phi(I^{\text{sec}}(i)), \\ I^{\text{sec}}(i+1) &= I^{\text{sec}}(i) \odot \exp\left(\alpha(\rho(I^{\text{cov}}(i+1)))\right) + \eta(I^{\text{cov}}(i+1)) \end{aligned} \tag{3}$$

where $\alpha(\cdot)$ is a sigmoid function scaled by a constant factor for clamping purposes, and $\odot$ denotes element-wise multiplication. The functions $\rho(\cdot)$, $\phi(\cdot)$, and $\eta(\cdot)$ are implemented using dense blocks, which are widely adopted for their strong representational capacity [4]. After the final conceal block, we obtain the stego image I^{steg} along with the lost information. Here each conceal block applies reversible transformations: the update to the cover image is an additive function, and the secret image transformation combines element-wise operations (exponentiation and addition) with functions that can be inverted using the same network structure in reverse. In addition, sharing parameters between forward and reverse passes ensures symmetry for perfect reconstruction. As a result, our i-Net is invertible, which enables the model to embed rich facial details into a visually natural frame, thereby preserving the semantic fidelity essential for downstream ADA task. In particular, our invertible design not only ensures **lossless reconstruction** where facial details—such as expressions, gaze, and posture—are fully recoverable, preserving both visual and semantic fidelity, but also **minimal perceptual distortion** where the generated stego frames remain indistinguishable from the original cover frames. Consequently, the standard invertible network training [10] used in this paper jointly optimize the embedding and extraction processes to maximize hiding capacity while preserving visual similarity between the stego and cover images.

Depression face video conceal process: In this stage, the model operates in the forward propagation mode of the invertible network, taking I_i^{sec} and I_i^{cov} as inputs. These inputs are passed through multiple hidden blocks to undergo feature transformations, embedding the semantic information of the secret image into the cover image while maintaining a visual appearance that closely resembles the original cover. The final output channels of the last hidden block produce the stego image I_i^{steg} along with the Lost Information. The resulting stego image I_i^{steg} visually approximates the original cover image while implicitly containing the embedded facial features. The optimization objective at this stage is to minimize perceptual distortion, ensuring that the secret image remains imperceptible to the human eye.

Depression face video reveal process: The decoding process runs the same network architecture in reverse. Taking the stego image I_i^{steg} and an auxiliary variable z sampled from a Gaussian distribution as inputs, the model passes them through a series of reveal blocks to reconstruct the original image pair $(I_i^{\text{cov}}, I_i^{\text{sec}})$. Owing to the strictly invertible transformations, the recovered I_i^{sec} achieves high fidelity at both pixel and structural levels, preserving fine-grained facial attributes such as expressions, gaze, and posture—thus providing a reliable foundation for subsequent depression analysis.

3. Experiment

3.1. Experimental settings

Datasets: We adopt the AVEC 2014 face video depression assessment dataset to evaluate the effectiveness of our proposed depression face video steganography method. It involves two types of recording conditions: Northwind and Freeform, each of which records 150 clips ranging from 6 seconds to 4 mins 8 seconds. Both datasets were evenly divided into three subsets: training, development and testing.

Metrics: We adopt two quantitative metrics to evaluate the quality of both cover/stego video pairs and secret/recovery video pairs: Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM). PSNR is used to measure the pixel-level fidelity between corresponding frames, while SSIM evaluates structural similarity and perceptual quality. Higher values of PSNR and SSIM indicate better visual quality and more faithful reconstruction. To further assess the semantic fidelity of the recovered video in the context of ADA, we follow [23] and employ a set of evaluation metrics including Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Pearson Correlation Coefficient (PCC), and Concordance Correlation Coefficient (CCC). These metrics jointly capture the regression accuracy and consistency between predicted and ground-truth depression scores.

Implementation details: Each video from the AVEC 2014 dataset is used as a secret video input. The secret video is first decomposed into individual frames, which are then one-to-one aligned with the frames of a generated cover video. Each secret frame is embedded into its corresponding cover frame using the reversible steganographic network. The resulting stego frames are then reassembled into a stego video, and we compute PSNR and SSIM to evaluate the quality of the hiding process. Next, the stego frames are passed through the decoding phase of the reversible network to reveal the recovered secret frames, which are subsequently reassembled into the recovered secret video. This recovered video is then fed into the ADA evaluation pipeline. The affective analysis results are compared with those obtained from the original AVEC2014 videos to assess the semantic fidelity of the recovered content.

3.2. Face video steganography and recover results

Table 1. Quantitative Evaluation of Steganographic Video Quality

Video Pair	**PSNR ↑**	**SSIM ↑**
Cover/Stego	49.82 dB	0.9939
Cover/Stego(LSB)	33.89 dB	0.9486
Secret/Recovery	48.88 dB	0.9956
Secret/Recovery(LSB)	32.05 dB	0.9120

To further validate the performance of our framework, we compare it with the classical Least Significant Bit (LSB) method, a widely used baseline in traditional steganography. As reported in Table 1, we evaluate both cover/stego and secret/recovery video pairs using PSNR and SSIM metrics. Our method achieves a PSNR of 49.82 dB and a SSIM of 0.9939 for the cover/stego pair, indicating minimal visual distortion. For the secret/recovery pair, it preserves fine-grained facial details crucial for ADA, significantly outperforming the LSB method in terms of fidelity and structure. These results demonstrate the advantage of our approach in achieving high imperceptibility while maintaining semantic integrity. This performance gain can be attributed to the integration of the invertible network module and the customized diffusion-based cover video generation, which together enable faithful concealment and accurate recovery.

3.3. Depression assessment results

Table 2. ADA evaluation metrics of original and stego videos, where Ff denotes the Freeform task; Nw represents the Northwind task; and Cb represents their combination.

Metric	RMSE	MAE	PCC	CCC
Original Cb	6.6222	5.4473	0.8170	0.7988
Steg Cb	6.6416	5.4550	0.8158	0.7976
Original Ff	6.6796	5.4529	0.8136	0.7957
Steg Ff	6.6946	5.4633	0.8127	0.7947
Original Nw	6.6963	5.4523	0.8123	0.7955
Steg Nw	6.7176	5.4605	0.8110	0.7943

Table 2 compares the results obtained from performing ADA on both the original dataset videos and the recovered videos after being steganographically encoded with our DFS framework. We analyzed the results for two types of recording conditions: Northwind and Freeform, and also computed the combined values. As shown in the table, the recovered videos after DFS encoding and restoration maintain RMSE, MAE, PCC, and CCC values that are close to those of the original dataset. Specifically, both RMSE and MAE show slight increases, with the maximum rise not exceeding 0.3%, while PCC and ACC experience slight decreases, with the maximum drop not exceeding 2%. These results suggest that our video steganography framework achieves high fidelity in preserving the affective characteristics of the original videos. The recovered videos remain suitable for ADA tasks, with negligible impact on affective evaluation performance, demonstrating the robustness and semantic consistency of our approach.

4. Conclusion

This paper proposes the first ADA-oriented face video steganography framework that leverages a reversible hiding mechanism and controllable cover video generation, which achieves high-capacity and high-fidelity information hiding with enhanced imperceptibility and enables flexible, prompt-driven control over the cover content. Extensive experiments demonstrated that our framework achieves strong performance in terms of PSNR and SSIM, indicating minimal visual distortion in both stego and recovered videos, validating the semantic integrity of our approach.

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